

AUDITING MODEL SUBSTITUTION IN COMMERCIAL LLM APIS

Are You Getting What You Pay For?

Software-only detection is unreliable under realistic deployment conditions; trusted hardware attestation provides an actionable path to verifiable inference.

• ML researchers

• AI security

• Cloud infra

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

COMMON SUBSTITUTION TYPES

1) Smaller model

Serve an 8B-class model when users pay for 70B-class inference.

2) Quantized variant

Swap FP16 weights for FP8/INT8 to cut GPU cost while keeping outputs similar.

3) Fine-tuned / updated

Ship an unannounced fine-tune or "silent update" (distribution drift).

4) Different family

Route to another architecture with a compatible interface (API black box).

FORMAL FRAMING

Auditing reduces to hypothesis testing over conditional output distributions.

● H_0

$$P_{\text{actual}}(y|x) = P_{\text{spec}}(y|x)$$

API matches the advertised model distribution.

● $H_1 \ P_{\text{actual}}(y|x) \neq P_{\text{spec}}(y|x)$

Provider swaps models (quantized/smaller/different family).

EMPIRICAL EVIDENCE

FP8 Quantization Preserves Most Accuracy – Making Substitution Hard to Detect

Across five model families, FP8 scores often fall within the original error bars (especially on MMLU and GSM8K).

Model family	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct vs FP8	62.69 ± 0.18 62.43 ± 0.26	61.14 ± 3.47 60.90 ± 4.10	20.65 ± 3.43 14.91 ± 2.49	22.62 ± 0.22 20.14 ± 0.24
Llama-3-70B-Instruct vs FP8	78.05 ± 0.08 77.88 ± 0.13	88.06 ± 1.44 87.35 ± 1.24	35.69 ± 1.33 35.75 ± 1.16	29.60 ± 0.51 33.12 ± 0.30
Gemma-2-9b-it vs FP8	71.86 ± 0.08 71.92 ± 0.11	81.80 ± 1.35 79.41 ± 1.14	33.41 ± 0.28 32.53 ± 0.34	28.64 ± 2.97 27.81 ± 3.22
Qwen2-72B-Instruct vs FP8	82.18 ± 0.08 81.98 ± 0.08	86.72 ± 1.00 86.82 ± 0.97	37.39 ± 1.41 37.67 ± 1.39	29.93 ± 2.71 31.08 ± 1.96
Mistral-7B-Instruct-v0.3 vs FP8	59.15 ± 0.10 58.77 ± 0.13	35.90 ± 4.54 32.20 ± 4.02	8.94 ± 1.24 7.68 ± 1.23	21.60 ± 0.17 22.72 ± 0.19

WHY HIGHER TEMPERATURE DOESN'T SAVE YOU

Higher Temperature Amplifies Performance Gaps but Increases Noise

 Benchmark accuracy vs temperature for five models across four tasks

Figure 1: Benchmark accuracy vs temperature (0.2, 0.5, 1.0, 2.0) across five models and four tasks.

OBSERVED PATTERN

At higher temperatures (e.g., 2.0), benchmark performance degrades across the board.

The gap between original vs substitute can narrow or become noisy – undermining text-output statistical tests.

IMPLICATION FOR AUDITING

If temperature increases output variance, then distinguishing $P_{\text{spec}}(y|x)$ vs $P_{\text{actual}}(y|x)$ requires more samples – not fewer.

Higher temperature can make the substitution signal harder to isolate.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

 Log probability variation across inference frameworks and hardware for Llama-3-8B

Figure 5: Same Llama-3-8B model yields different log-probabilities across software versions (transformers, vLLM) and hardware (A100, H100).

ROOT CAUSE

Production inference introduces nondeterminism: frameworks and GPU kernels change numerical behavior.

WHY THIS DEFEATS FINGERPRINTING

Auditor compares token-level $\text{logp}(x,y)$.
But logp varies even when the underlying model is identical.

An auditor cannot distinguish “same model on different infra” from “substitute model” using log-prob metadata alone.

Result: unreliable decisions and false accusations.

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

ATTACK STRATEGY

Provider mixes a cheaper substitute response with probability p . As $p \rightarrow 0$, the mixed distribution approaches the specified model.

$$P_{\text{mix}}(y|x) = (1 - p) \cdot P_{\text{spec}}(y|x) + p \cdot P_{\text{sub}}(y|x)$$

Small $p \Rightarrow$ tiny divergence $\Rightarrow$ many more samples needed to reject H_0 .

INTUITION

Text-output tests (e.g., MMD) are sample-hungry. Randomization makes them effectively query-prohibitive.

Low-rate substitution

Most outputs match $P_{\text{spec}} \rightarrow$ auditor sees near-normal behavior.

Detection cost explodes

To confidently detect, auditor must spend large query budgets (time + money) and still faces noise.

Benchmark evasion (related)

Route known benchmark prompts to the real model; substitute the rest.

Software-Only Auditing Methods Are Fundamentally Unreliable

Across realistic deployment conditions, each software-only strategy has a structural failure mode.

METHOD TYPE

Text-based statistical tests
E.g., MMD on sampled outputs.

Log-probability methods
Token-level logp “fingerprints”.

Adversarial countermeasures
Benchmark evasion, randomized routing.

WEAKNESS

Small effect sizes
FP8 quantization can preserve benchmark behavior within error bars.

Nondeterminism
Logp varies across inference stacks and hardware even for the same model.

Adaptive attacker
Provider controls routing + prompt recognition in the serving stack.

PRACTICAL LIMITATION

Query budget
Needs impractically many API queries; high temperature adds noise.

False positives
Cannot separate infra drift from substitution in production.

Query-prohibitive
As substitution rate decreases, distributions converge and detection cost explodes.



ACTIONABLE ALTERNATIVE

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

WHAT TEES GIVE YOU

Provable

Cryptographic attestation that specific inference code + model weights are loaded and executed.

Efficient

Only modest performance overhead versus heavyweight alternatives (e.g., ZKPs for large models).

Robust

Not fragile to temperature, sampling variance, or infrastructure nondeterminism.

Core idea: verify execution, not outputs.

PRACTICAL DEPLOYMENT PATH

- 1) Run inference inside a hardware-secured enclave
- 2) Produce an attestation report (code + weights)
- 3) Auditor verifies the report before trusting responses

Example: NVIDIA confidential computing on H100 GPUs enables TEE-like workflows for modern inference stacks.

Contrast: software tests can be evaded; attestation binds claims to hardware guarantees.

KEY TAKEAWAYS

Close the Verification Gap with Hardware Attestation

- 1) Model substitution is economically motivated and hard to audit via black-box access.
- 2) FP8 quantization can barely affect benchmarks – often within error bars – making statistical detection unreliable.
- 3) Inference nondeterminism across frameworks/hardware defeats log-probability fingerprinting.
- 4) Randomized substitution and benchmark evasion make detection query-prohibitive.
- 5) TEEs are the most actionable path today for provable integrity guarantees in LLM APIs.

● Call to action: standardize attestation for verifiable inference.